Department of Information Technology
Academic Year 2024 - 2025 (Even Semester)

Degree, Semester & Branch: B. Tech, VI & IT

Course Code & Title: CCS335 & Cloud Computing

Name of the Faculty member: A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic: Unit IV/ OpenStack**
- **Course Outcome:** CO4
- **Activity Chosen:** Think-Pair-Share (TPS)
- **Justification:**
  In Unit IV, understanding the concepts and components of OpenStack is essential for students. The Think–Pair–Share activity supports collaborative learning, encouraging students to first think independently, then discuss ideas with a partner, and finally share their understanding with the class. This method enhances critical thinking, communication, and teamwork, all vital skills in cloud computing.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
  The Think–Pair–Share activity on the topic *OpenStack* was carried out in a structured and engaging manner. Initially, the topic was introduced to the students, setting the context for discussion (Figure 1). The class was then divided into 28 pairs, encouraging peer interaction and collaborative thinking. Each pair was given approximately five minutes to discuss and share their understanding of the topic with one another. This allowed students to process the information individually and refine their ideas through dialogue. Following the pair discussions, the class entered the sharing phase, where selected pairs were invited to present their insights and interpretations to the entire class (Figure 2). Among the presenters, the team led by Ms. Shanmuga Priya V stood out by delivering a clear, structured, and comprehensive explanation of OpenStack and its core components. This activity not only deepened the students' understanding of the topic but also enhanced their communication and teamwork skills.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO9	P010	P011	P012	PSO3
CO4	3	2	3	2	2	3	3	3

(1-Low 2-Mederate 3-High)

Innovative practice	PO1	PO2	PO5	PO9	P010	P011	P012	PSO3
	3	2	3	2	2	3	3	3
Justification for correlation	Students apply knowledge of IT and networking protocols to create cloud services.	Students identify OpenStack's functionalities and computational capabilities.	Learners evaluate tools and simulations for cloud environment setup.	Students demonstrate communication and teamwork through discussion and presentations.	Students interpret both technical and non-technical content effectively.	Learners understand steps involved in setting up cloud infrastructure and deploying services.	Students recognize the need for continuous learning in cloud computing technologies.	Students strengthen critical thinking to anticipate outcomes and propose solutions in cloud environments.

- **PO / PSO mapped:**
- **Images / Screenshot of the practice:**

Figure 1. Discussion on the topic Global State snapshot Recording

Screen shots

Figure 2. Shanmuga Priya V Presentation about OpenStack

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

- Students actively participated in the activity.
- Many expressed that the activity improved their ability to recall concepts and communicate effectively.

❖ ***Benefit of the practice:***

- Students gained a better understanding of OpenStack.
- The interactive nature of the activity promoted effective peer-to-peer and student-faculty communication.
- Students felt more comfortable sharing their ideas with peers before speaking to the whole class.

❖ ***Challenges faced in implementation:***

- Some students hesitated to present in front of the class.
- Although the session was planned for 10 minutes, it extended to 15 minutes due to delays in student participation.

References:

- https://www.youtube.com/watch?v=CegPJytwsqE
- https://www.youtube.com/watch?v=moSWS1Gha2g
- https://www.youtube.com/watch?reload=9&v=in5-pwSVxwc
- https://www.youtube.com/watch?v=1yRJf8j8xVU

Signature of Faculty Member **HOD**